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THEORETICAL ARTICLE

Q-OPTBERT: a quantum-optimized DistilBERT model for sentiment analysis using QT-TSA

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Abstract

This research presents a new hybrid quantum deep learning model called Q-OptBERT aimed at accomplishing sentiment analysis through quantum optimization. The solution combines DistilBERT to extract contextual embedding with a Variational Quantum Circuit (VQC) capable of capturing nonlinear linguistic dependencies by utilizing quantum superposition and entanglement as principles of quantum physics. A new Adaptive Quantum Feature Reduction (AQFR) module will allow for dimensional reduction while preserving semantic richness in embedding space. The primary contribution of the research is the introduction of the Quantum Tuned Tunicate Swarm Algorithm (QT-TSA), a biologically-inspired optimization approach to enable faster convergence and diversity of solutions when tuning hyperparameters. This novel optimization process differs fundamentally from other optimization methods such as Particle Swarm Optimization (PSO) or Quantum Genetic Algorithm (QGA) because QT-TSA can dynamically assess and update its own search parameters through the use of quantum potential functions to robustly and systematically explore parameter space. The combined quantum swim method creates an optimization strategy that optimizes the learning rate and entanglement strength of the VQC layer. The experimental results on relevant benchmark datasets show superior accuracy, precision, and less computational time than other Bidirectional Encoder Representations from Transformers (BERT) or hybrid quantum NLP models. Q-OptBERT is a promising next-generation hybrid quantum deep learning model for effective, comprehensible, and high-performance quantum-enhanced natural language processing because of its unique blend of quantum circuit meta-learning, feature adaptation, and swarm optimization.

Keywords Sentiment analysis · Q-OptBERT · VQC · AQFR · QT-TSA · DistilBERT

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1 Introduction

However, traditional deep-learning and transformer-based methodologies have many inherent limitations as well. The identified issues experienced from these methodologies include: extremely high effects on computing resources, the need for a very large quantity of labelled training datasets, and scalability problems associated with using them for real-time and streaming environments [1, 2]. Additionally, as the depth of layers within a model increase, and the total number of parameters associated with each layer increase, the time required to train a model will proportionately increase, and the overall complexity of optimizing a model will also increase [3, 4]. Thus, from those two factors, convergence rates will decrease, and models will have greater opportunities to reach "local minimum" points. Classical neural algorithms, while improved by means of attention mechanisms, are still limited in the number and complexity of emotions, feelings, and opinions they can identify with a nuanced understanding of higher-order nonlinear relationship and long-range semantic correlation across linguistic ambiguities or changes within sentiment [5, 6]. Quantum computing is a relatively new approach to computing which may be used to solve the problems described above [7].

Quantum computers operate differently from classical computers, which store data as binary digits (0s and 1s) [8, 9]. Instead of binary digits, quantum computers store data using qubits that leverage the principles of quantum mechanics through such principles as superposition, entanglement, and interference. By utilizing these properties, the quantum system can represent and manipulate data in Hilbert spaces of size exponentially larger than classical computers and can perform many of the mathematical operations needed to model high-dimensional data (e.g., data from several different types of features) simultaneously; such operations would normally require a substantial amount of time if performed by classical computer systems [10, 11]. For this reason, quantum machine learning and quantum-enhanced natural language processing are receiving increasing attention as potential approaches to the high-dimensional language understanding tasks of today's world. Building on this work, researchers have shown that recent quantum natural language processing studies have indicated that hybrid quantum-classical systems can benefit in terms of representation expressiveness and computational efficiency with the combination of classical neural networks and variational quantum circuits [12, 13]. Although these hybrid quantum-classical systems have been beneficial in a few instances, the current models of quantum-enhanced sentiment analysis have a number of disadvantages, including inefficient encoding of the feature representations, lack of scaling to larger data sets, high susceptibility to noise on near-term quantum devices, and suboptimal hyperparameter optimization methods [14]. In particular, mapping the high-dimensional contextual embeddings from large language models (such as BERT) to quantum circuits results in a very high requirement for qubits and a large depth of quantum circuits, thus creating serious limitations for practical use [15].

1.1 Background and motivation

The combination of Quantum Computing and NLP is paving the way to create novel methods of addressing complex language processing tasks for the future. Traditional NLP models rely on either deterministic or probabilistic (numeric) optimization in Euclidean space. This creates limitations on how effectively these models can be applied to represent the complex nature of language semantics as the relationship becomes increasingly multidimensional and intertwined. Language is characterized by the fact that its meaning may differ depending upon the context in which it is used. Language also contains an inherent amount of ambiguity. Both the nonlinearity and the ambiguity of language correlate with the inherent probabilistic and superpositional characteristics of quantum systems. Quantum-inspired or quantum-native models of NLP use the properties of quantum systems to encode linguistic features as quantum states and encode semantic interactions as parameterized circuits on these states. VQC allow for flexible hybrid learning by combining quantum state evolution with classical approaches to optimization using looped calculations. However, these types of models are still in the early stages of development and there are several barriers that hinder their adoption in practice including issues related to the high-dimensional space of language embeddings, noise sensitivity, and the absence of reliable optimization techniques specifically designed for hybrid quantum/classical model structures.

A significant shortcoming of most current quantum NLP models is the reliance on traditional techniques for hyperparameter tuning. Most of the existing quantum NLP models were trained using standard optimization methods (i.e., PSO, GA, grid-based searches). These methods frequently have inadequate adaptability to the dynamic loss landscapes provided by quantum circuits, limited exploration capability, and premature convergence. Therefore, in hybrid quantum NLP systems, obtaining steady and effective training continues to be a bottleneck. Inspired by these difficulties, this work investigates how to combine quantum-aware swarm optimization and adaptive feature reduction into a single sentiment analysis system. The approach lowers quantum resource requirements without compromising semantic expressiveness by implementing AQFR. Concurrently, a quantum potential-guided search mechanism that improves exploration, steers clear of local optima, and speeds up convergence is introduced by the suggested QT-TSA. When combined, these elements allow for a quantum-enhanced sentiment analysis model that is more useful, scalable, and comprehensible.

The contributions of this paper are manifested below,

- This work introduces Q-OptBERT, a novel hybrid quantum–deep learning model that integrates DistilBERT with a VQC to capture complex nonlinear linguistic dependencies using quantum superposition and entanglement.
- This work proposes an AQFR module, which minimizes feature redundancy while preserving key semantic information, enabling efficient dimensionality reduction in quantum-compatible embeddings.
- This work develops QT-TSA a new bio-inspired optimization technique that dynamically adapts search parameters through quantum potential functions to

enhance convergence speed and solution diversity during hyperparameter optimization.
- This work optimizes key model parameters, including the learning rate and entanglement strength of the VQC layer, through the hybrid quantum–swarm optimization mechanism, achieving balanced performance between exploration and exploitation.

For the remainder of this paper, it is organized as follows. Section 2 offers a problem statement and a list of relevant works. Section 3 presents and explains the suggested protocol. After that, the Sect. 4 presents the findings and discussion, and the Sect. 5 offers the conclusion.

2 Literature review

In 2025, Kayal et al. [16] proposed a Quantum Natural Language Processing (QNLP) framework that fuses the principles of quantum computing with natural language processing frameworks to push forward the work of sentiment analysis. The framework adds to the accuracy of QNLP by combining quantum-enhanced feature extraction and classification mechanism, with some degree of interpretability. In considering QNLP a superior potential over classical SA systems, it also covered the continued challenges related to scalability and implementation and recognized that this initial research would promote use of researching sentient via quantum.

In 2025, Dave et al. [17] focused on improving sentiment analysis through a quantum-fuzzy neural network (QFNN) model that pairs principles from quantum computation with fuzzy logic. The authors hail QFNN an improvement over classical NLP models in its ability to manage problems with noisy data, uncertain linguistic structures, and high-dimensional text through quantum-enhanced fuzzy reasoning principles.

In the year 2023, Ruskanda et al. [18] researched how quantum computing can be utilized with Natural Language Processing (NLP) tools for good sentiment classification. Researchers presented a Quantum-Enhanced Support Vector Machine (QE-SVM) framework exploiting quantum computing's largest benefits of quantum parallelization and superposition properties to perform more mathematically complex computations at faster speeds. The model trains circuit parameters and converts the original data through a multi-staged process-text to circuit conversion, quantum parameter training, state vector generation, and classification.

In 2024, Chu et al. [19], introduce Complex-valued Quantum-enhanced Long Short-term Memory Network (CQLSTM) which is a quantum-inspired, complex-valued, sentiment analysis approach that introduces complex-valued embeddings which are based on density matrices in order to create richer semantic correlations between word embeddings. Additionally, the model, taking advantage of the quantum measurement approach, encodes contextual dependencies using the quantum-enhanced LSTM module that is able to capture long-term relationships amongst the emotional features.

In 2023, Lai et al. [20] presented a Quantum-inspired approach to Natural Language Processing, namely, complex-valued neural networks with quantum operations. The complex-valued neural network (ComplexQNN) approach integrates a complex-valued embedding layer, a quantum encoding layer, and a measurement layer that aligns NLP representation with Hilbert space theory. Distinctions from conventional models include that ComplexQNN method integrates the imaginary components of the complex-valued representation to reveal latent semantic information through additional expressive power.

In 2025, Singh et al. [21] examined Quantum Neural Networks (QNN) in order to analyze sentiment, emotionality, and sarcasm in multimodal dialogues. The use of the Variational Quantum Eigensolver (VQE)—a hybrid quantum-classical algorithm—allowed the authors to iteratively update the quantum neural network's parameters via optimization while simultaneously minimizing Hamiltonian eigenvalues. The hybrid QNN provided an improved modeling of the complexities of intensity and dependencies across modalities compared to deep learning paradigms while also effectively unrolling the particulars of decoding human expression types or shades of feeling. Simply put, the model significantly improved overall multimodal sentiment classification.

In 2024, Phukan et al. [22] proposed a combined quantum model, which consisted of a Variational Quantum Circuit (VQC) and a dense neural network to collaboratively identify sarcasm, sentiment, and emotion across multimodal input data. The researchers utilized principles from quantum mechanics (superposition, entanglement, and interference) to capture complex relationships across text, auditory, and visual modalities complemented by their crude sentiment detection model.

In 2024 Ravikumar et al. [23] developed Quantum-Accelerated Hyperparameter Tuning (QAHT), a quantum-enhanced framework for efficiently tuning large NLP models at scale. By adopting a Quantum Neural Network (QNN) to adaptively tune hyperparameters "on the fly", QAHT adapts to the streaming data input, adapting the model architecture in real-time, subsequent to the CNC adaptation method. Unlike traditional grid or random hyperparameter formulation, the framework used quantum parallelism to observe in pretty much real-time to produce actionable hyperparameter tuning and get feedback on improvements in learning. In all, QAHT was designed to improve the learning efficiency and adaptability for a sentiment model while insisting on model stability across differing text environments.

In 2025, Li et al. [24] introduced an innovative hybrid quantum-classical design known as the Quantum Multimodal Learning for Sentiment Classification (QMLSC) model that unifies text and speech data to represent sentiments more robustly. This model utilizes optimized Variational Quantum Circuits (VQCs) and also residual fusion structures between the quantum layers and classical layers in order to obtain richer sentiments and minimize the loss of information. Unlike classical architecture, QMLSC can utilize randomized expressive circuits for improved flexibility, robustness, and accuracy.

In 2025, Xu et al. [25] also introduced the quantum-inspired Aspect-Based Sentiment Analysis (ABSA) model to help integrate complex semantic dependencies in long text sequences more efficiently. Our paper focused on proposing an ABSA method, which we called a quantum embedding module, that would embed the word

features into superposition multiple states in a complex-valued Hilbert space in order to achieve a more enriched semantic representation. Furthermore, we added a quantum cross-attention mechanism to integrate the aspect opinion dependencies to provide awareness of contextual awareness.

2.1 Problem statement

Sentiment analysis is a core activity in NLP - by finding sentiments and opinions contained in the text data. Traditionally, however, model frameworks using deep learning and transformer-based methods are computationally costly, affected by gradient vanishing, and limited in their effectiveness of comprehensively utilizing sophisticated contextual dependencies, especially in longer or ambiguous sentences. Moreover, they are highly dependent on very large labelled datasets, and do not employ techniques to help with the interpretability of sentiment reasoning and contextual dependencies. Integrating quantum optimization techniques in the realms of quantum information and computation, with NLP present a compelling path to tackling these shortcomings. Richer semantic linkages and quicker convergence during sentiment classification are made possible by probabilistic interference and parallel state representation made possible by quantum-inspired computation. However, there is still much to learn about the best way to create and train such quantum-enhanced models. In order to advance scalable, interpretable, and high-performance sentiment understanding in complex textual environments, this research attempts to create a framework that effectively captures contextual nuances, increases accuracy, and lowers computational overhead.

3 Proposed methodology

3.1 Proposed workflow steps for Q-OptBERT

In artificial intelligence, NLP is an area that is centered on deriving contextual and intent-based emotions from language. Emotion and opinion detection are more common descriptors than sentiment analysis as we think of emotion connected to feelings of joy, sadness, or any other spectrum of human emotion. In analyzing sentiment as it relates to language, many of the deep learning layers used to extract emotions in textual data have had some ongoing issues or shortcomings. These concerns include but are not limited to high computational costs and training times, insufficient resolution or context, and slow convergence rates. These issues stem from the complexities of the features of linguistic structure, and representation models require large amounts of data. These shortcomings can largely describe the current limitations of artificial intelligence in NLP. While we will not solve all of these issues, we will introduce the Q-OptBERT framework which uses quantum-inspired optimization to address these short comings of pre-trained language models and efficiently provide better solutions by using multi-dimensional understanding.

3.1.1 Preprocessing

The preprocessing step is often the first stage of processing raw textual data for use within the Q-OptBERT framework sentiment analysis. Because natural language data often has inconsistencies, inaccurate or excessive symbols, and noise; pre-processing into a form that allows methodological standardization of the inputs, and sematic coherence, is important prep and preprocessing generates the inputs that will capture and define the embedding. Preprocessing in the classical sense will contain several NLP tasks including tokenization, lowercasing, stop-word removal, punctuation filtering, and text normalization.

- Tokenization

Tokenization is the process of splitting a text into smaller units called tokens, which can be words, sub-words, or characters. For example, the sentence "Quantum computing enhances NLP models" is decomposed into the tokens ["Quantum", "computing", "enhances", "NLP", "models"] so that analysis of the linguistic structure can take place and numerical embedding can be produced for each token. In Q-OptBERT we use a tokenizer compatible with DistilBERT to ensure an identical sub-word segmentation, which is applied to the vocabulary of the pretrained embedding.

- Lowercasing

Lowercasing is the process of converting all text to lowercase characters so that uniformity can be produced and vocabulary sparsity can be reduced. After lowercasing, for example, "Quantum" and "quantum" are considered to be identical tokens and therefore there will not be two embeddings of the same representation, represented in the embedding layer. This process helps to reduce noise at the embedding layer while increasing computational efficiency and minimising interference of the original contextual integrity in the embeddings. The proposed architecture is illustrated in Fig. 1.

- Stop-word removal

Stop-words are high frequency words (such as "the," "is," "an," and "of") that generally contribute little to the semantic validity of the sentence in the case of sentiment classification. By removing them, we can reduce the input sequence length and focus the model on tokens carrying sentiment. A list of stop-words included within the domain was created taking into consideration contextually significant modifier terms (e.g., "not," "never") which can influence polarity of sentiment detected.

- Punctuation filtering

Punctuation marks and special characters are stripped from the words in order to keep the embedding representation free of extraneous noise. However, marks that denote meaning, such as an exclamation mark or question mark, which typically indicate

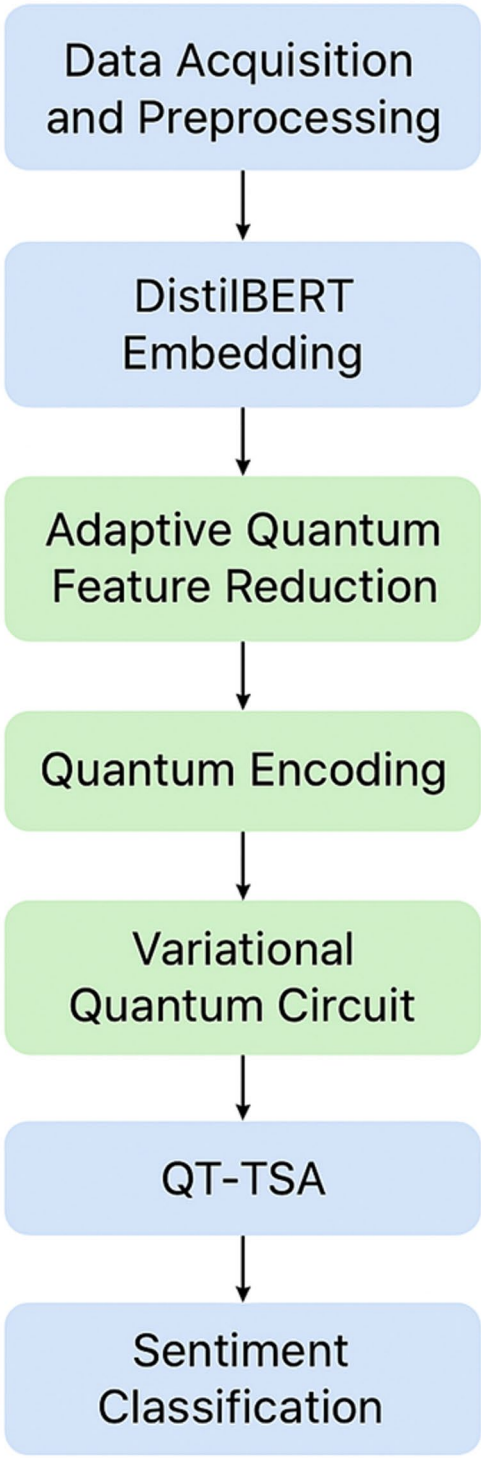


Fig. 1 Overall proposed architecture

increased emotional tone, are intentionally retained. This allows us to filter out noise, while preserving cues for sentiment that exist in punctuation.

- Text normalization (Stemming and Lemmatization)

Text normalization is a type of word standardization that lowers morphological complexity by reducing some words to their base or root form, which can help reduce lexical diversity. Stemming is an approach to standardization that extracts the stem of the word by removing suffixes, for example: "computational" becomes "comput." Stemming leads to computationally less intensive embedding, but as the original meaning is disrupted, the term may not be conceptually sound. Lemmatization captures these lexical and morphological relationships by leveraging linguistic rules of a language and exists in lexical databases. For example, during lemmatization, "better" would resolve to its canonical form "good". In Q-OptBERT, lemmatization is any enhancement of meaning that is appropriate for the purpose of successful context-aware embeddings through DistilBERT.

With these preprocessing tasks in place, the text is represented in a simple standard semantic form that is ready for quantum functional embeddings using feature encoding. The first stage of multi-stage processing reduces redundancy in the data, increasing its quality as a feature system that can incorporate context-aware representations in the subsequent Q-OptBERT context embedding process.

3.1.2 Contextual text embedding using distilbert

After the data is preprocessed, the cleaned and normalized textual data is embedded into high-density numerical vectors through the binding DistilBERT model. This embedding process is an essential step to capture the semantics and syntax of language used to understand contextual relationships in the sentence. The second phase of the Q-OptBERT framework utilizes the DistilBERT model to embed the preprocessed textual data into dense, semantically relevant vector representations. This embedding process is critical for capturing the contextual and syntactical dependencies that support a proper understanding of sentiments. DistilBERT was developed through a knowledge distillation process from a larger original "teacher" model (BERT), in which a smaller "student" model learns to apply the same representational capacity as the larger original model. This approach provides access to a model that offers 95% of BERT's performance while being 40% smaller and 60% faster than BERT, and more computationally efficient for the hybrid quantum-classical system. The model relies upon multiple transformer encoder layers to model bidirectional dependencies of the input sequence while ensuring that the representation of each token is informed by its full context in the entirety of the sentence.

Each input sentence is processed through a WordPiece tokenizer which breaks the sentence apart into a sequence of tokens. These tokens are then embedded into numerical vectors using token embeddings depending on the positional and segment embeddings Eq. (1).

$$E_i = E_{token}(x_i) + E_{position}(i) + E_{segment}(s_i) \tag{1}$$

Where, $E_{token}(x_i)$ represents the learned embedding of the ith token, $E_{position}(i)$ encodes the positional index, and $E_{segment}(s_i)$ distinguishes between sentence pairs when applicable. This merged embedding E_i is subsequently inputted into the transformer encoder layers for contextual modeling.

The primary benefit of DistilBERT is its multi-head self-attention mechanism, which captures complex inter-token associations. The self-attention operation is shown in Eq. (2).

$$attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{2}$$

DistilBERT's efficient architecture and lower number of parameters make for a suitable modeling option for quantum integration. Besides being of high-quality contextual embeddings, the low-level computation required reduces the complexity of the downstream quantum encoding and optimization processes. Lastly, the embeddings outputs from this component will serve as inputs for the AQFR module to minimize redundancy in dimension without sacrificing semantic richness or meaning.

3.1.3 AQFR

The process of reducing embeddings will be performed while trying to ensure that important semantic contextual information is not irreversibly lost. AQFR module performs the aforementioned embeddings process through its quantum-inspired evaluation of dimensional entropy and its application of adaptive thresholding measures to select which embeddings output features are more important. The AQFR module seeks to reduce output redundancy and dimensions of output embeddings while conserving sentiment-related meaning. Rather than typical feature selection processes that rely on linear correlations or outcomes that may have low variance, AQFR will utilize quantum-aware entropy-based variability weighting and quantifying probabilistic value of relationships between attributes of embedding dimensions to determine desired certainty or contribution to general context appropriated for sentiment. This approach allows for arriving at reasonable compromise between compression of embeddings in gaze of retention of richness of scales of meaning especially in reference to compressed schema of embedding restructuring into quantum-based states such that prescriptive encoding of a presented data set occurs more optimally.

3.1.3.1 ***Quantum-inspired entropy evaluation*** A more formal approach to measure embedding dimension is to consider each embedding dimension as a feature with probabilistic character whose overall contribution sentiment representation can be

cooperative measured by quantum inspired entropy. The entropy of a feature f_i is calculated according to Eq. (3).

$$H\left(f_i\right) = -\sum_{j=1}^{n} p_{ij} \log_2\left(p_{ij}\right) \tag{3}$$

Where p_{ij} is normalized probability representing amplitude of the ith feature across all samples. Higher feature entropy represents greater information diversity. Hence, greater entropy implies a feature likely carries some contextual variance that has value to classify sentiment.

3.1.3.2 ***Adaptive feature selection mechanism*** Following the calculation of entropy, AQFR utilizes an adaptive threshold function to select parameters that are informative.

$$F' = \{f_i \epsilon f | H\left(f_i\right) \geq \tau_{adaptive}\} \tag{4}$$

In this, F is the set of all original features, and $\tau_{adaptive}$ is a threshold that is adaptively devised from the mean and standard deviation of all feature entropies. Such adaptability enables the action of removing or retaining the selected features to be dictated by the complexity inherent in the input distribution, and prevents under- or over-compression.

3.1.3.3 ***Output and quantum compatibility*** The reduced embedding set F' serves as the compact, information dense vector space for quantum encoding, and these features are directly encoded into quantum states in the second phase of the VQC framework using amplitude or angle encoding. AQFR optimizes quantum resources, depth of the circuit, and training efficiency in classical-quantum multimodal adaptation because with the reduced embedding space of the trained DistilBERT embedding, semantic fidelity is optimal through the re of redundant dimensions; AQFR has been deemed essential in use for the transition from the classical DistilBERT embedding phase to the quantum encoding layer for the downstream task. The AQFR module assures that only the most discriminative and semantically-relevant features are preserved while being guided by entropic adaptive reduction, ensuring the compact embeddings can provide efficient and effective quantum state representation in the next portion of the Q-OptBERT architecture.

3.1.4 Quantum encoding and VQC design

After features have been reduced by the AQFR module, the compact, semantically rich embeddings are then mapped into quantum states to be processed using a VQC. This stage is the quantum element of the propose Q-OptBERT framework, taking

advantage of quantum parallelism and entanglement to learn more complex linguistic dependencies than classical neural architecture provide.

3.1.4.1 ***Quantum state encoding*** The reduced embedding vector $F' = [f_1, f_2, \ldots, f_m]$ obtained from AQFR is encoded into a quantum state using amplitude or angle encoding techniques. In amplitude encoding, the normalized embedding is represented as a superposition of computational basis states as per Eq. (5),

$$|\varphi\rangle = \sum_{i=1}^{m} f_i \, |i\rangle \, , \;\; with \sum_{i=1}^{m} |f_i|^2 = 1 \tag{5}$$

Here, $|\varphi\rangle$ denotes the encoded quantum state, $|i\rangle$ and represents the basis vector corresponding to the ith feature. This mapping allows all features to coexist in parallel within a single quantum register, offering exponential data representation efficiency.

3.1.4.2 ***VQC architecture*** The encoded state is processed through a parameterized quantum circuit comprising rotation and entanglement gates. Each layer of the VQC includes single-qubit rotation gates $R_y(\theta_i)$ and $R_z(\varnothing_i)$, followed by controlled entanglement operations (e.g., CNOT gates) that establish quantum correlations among qubits. The general transformation of a quantum layer can be expressed as per Eq. (6).

$$|\varphi'\rangle = U(\theta) \, |\varphi\rangle \tag{6}$$

3.1.4.3 ***Quantum entanglement for linguistic dependency modeling*** Due to quantum entanglement, relationships between a number of linguistic features, which are hard to model in classical models, can all be modelled at once. Entanglement sufficiently represents semantic interdependence, phrase level interactions, and contextual polarity shift in sentiment expressions with the ability for the change of one qubit to influence other qubit states. This features thus increases the ability of the model to generalize out of complex verbal constructs such as compound feelings, sarcasm, and simply negations.

3.1.4.4 ***Output representation*** The VQC's final output after quantum measurements will yield expectation values for sentiment-related features. These vectors will repre-

sent context-sensitive sentiment embeddings and are then passed on to the QT-TSA layer for parameters tuning and final classification.

3.1.5 Hybrid quantum-classical training architecture

At the very end, the winning result of the VQC, in fact of classically in the output of the quantum measurement after all, will be expectation values of sentiment features. These quantum-mapped vectors will be passed on into the QT-TSA layer for parameter tuning followed by their final classification.

3.1.5.1 ***Hybrid learning workflow*** The start of the hybrid training will be with DistilBERT to create contextual embeddings. These embeddings are then made smaller via the AQFR module. The compressed embeddings are then transformed into quantum states, which the VQC processes. The quantum measured output, or the expectation values, is then extracted from the quantum observables (like Pauli operators), and passed to the classical module, to account for the gradients. The mathematical representation for computing an individual observable O is as in Eq. (7).

$$\langle O \rangle = \langle \varphi' | \, O \, | \varphi' \rangle \tag{7}$$

Where $|\varphi'\rangle$ is the output state of the parameterized VQC after applying the unitary transformation $U(\theta)$. This value acts as a quantum feature that influences the loss function used during training.

3.1.5.2 ***Joint optimization mechanism*** In practice, the optimization process is a combination of classical backpropagation and quantum parameter updates, within a single training loop. The loss function L, is dependent on the classical parameters w (from DistilBERT and AQFR) and the quantum parameters θ (from the VQC). The overall gradient update can be displayed as in Eq. (8),

$$\nabla L = \frac{\partial L}{\partial w} + \frac{\partial L}{\partial \theta} \tag{8}$$

Here, $\partial L / \partial w$, is obtained through the back-propagation process, $\partial L / \partial \theta$, is determined using parameter-shift rules, and a method for quantum gradient estimation that circumvents the collapse of the wavefunction during observation.

3.1.5.3 ***Quantum-classical co-optimization*** This co-optimized framework offers a smooth interplay of quantum parallelism and classical convergent control as a hybrid system. Quantum modules use state superposition and entanglement to enable high-dimensional nonlinear mappings and enlarge the representation space, compared with classical modules that offer gradient space stability, batch learning efficiency, and compatibility with hardware. By maintaining the interactive exchange and evo-

lution between both subsystems, the hybrid model learns to represent complicated linguistic dependencies more effectively than the purely classical models, yielding an avoidable trade-off between accuracy, explainability, and compute expense onboard.

3.1.5.4 ***Outcome of hybrid training*** At convergence, the hybrid systems produce optimized parameters for both the encoder (DistilBERT) and the VQC model. The hybrid learning model achieves improved generalization and convergence efficiency, while providing robustness across a variety of sentiment datasets. Thus, the trained quantum-classical fusion becomes the main inference framework for Q-OptBERT for meta-heuristic tuning for sentiment prediction.

3.1.6 QT-TSA

The final optimization layer of the proposed Q-OptBERT framework employs a quantum Tunicate Swarm Algorithm (QT-TSA) for a hyperparameter tuning mechanism across both the classical DistilBERT and the quantum VQC components. QT-TSA expands the original Tunicate Swarm Algorithm (TSA), inspired by the socialized and jet-propelled foraging behavior of marine tunicates, with quantum potential dynamics to improve exploratory, converging and stability characteristics of the learning algorithm. TSA is based on the collective behavior of tunicates that use swarm coordination and jet propulsion to advance in the direction of prey. Three behavioral constraints are used by each tunicate (agent) to update its position:

- Avoiding collisions between neighboring agents,
- Moving toward the best-performing individual, and.
- Converging to the optimal region while sharing positional information.

The avoidance mechanism is represented as per Eq. (9).

$$\overrightarrow{a} = \frac{\overrightarrow{g}}{\overrightarrow{m}} \tag{9}$$

where $c_1,\ c_2, c_3 \in\ [0,1]$ are random coefficients, and $\overrightarrow{m}$ denotes the tunicate's social strength, given by Eq. (10).

$$\overrightarrow{m} = p_{min} + c_1 \cdot\ (p_{max} - p_{min}) \tag{10}$$

$$\text{With } p_{min} = 1 \text{ and } p_{max} = 4,$$

With $p_{min} = 1$ and $p_{max} = 4$,
After preventing collisions, each agent moves toward the best solution x_{best} using Eq. (11),

$$pd = |x_{best} - r_{rand} \cdot\ x\,(t)\,| \tag{11}$$

and the position update rule is:

$$x\left(t+1\right)=\begin{cases} x_{best}-\vec{a}.pd, & if\ r_{rand}<0.5 \\ x_{best}+\vec{a}.pd, & otherwise \end{cases} \tag{12}$$

This ensures adaptive convergence toward the global optimum. The swarm behavior is then modelled by positional sharing between agents using Eq. (13).

$$x_i\left(t+1\right)=\begin{cases} x_i\left(t\right)+\frac{x_{i-1}(t+1)}{2}+c_1x_i\left(t\right), & i>1 \\ x_i\left(t\right), & i=1 \end{cases} \tag{13}$$

- Quantum-tuned enhancement

In order to improve upon the limitations of local trapping and slow convergence in high-dimensional optimization of TSA, QT-TSA is proposed which employs quantum tuning parameters to alter the laws of the agent motion, namely, their probabilistic potential field. The quantum-inspired update according to Eq. (14) implies a tunnelling probability defined by a quantum factor (QF).

$$x_i^{t+1}=x_i^t+QF\cdot\ \sin(\omega\ \cdot\ r)\cdot\ (x_{best}^t-x_i^t) \tag{14}$$

where, $QF=\ \alpha\ e^{-\beta\ d_t}$ depends on the swarm diversity d_t, with α and β controlling the adaptive exploration–exploitation balance. This integration allows agents to probabilistically "tunnel" through potential barriers in the fitness landscape, ensuring more diverse and global search behavior.

- Fitness evaluation and objective function

Each agent's fitness is evaluated based on model accuracy, expressed as per Eq. (15),

$$fitness=\max\left(p\right),\ p=\frac{TP}{TP+FP} \tag{15}$$

Where TP and FP represent true and false positives, respectively. The best-performing solution x_{best} corresponds to the parameter configuration yielding the highest sentiment classification accuracy. In Q-OptBERT, QT-TSA optimizes both DistilBERT learning rates and VQC entanglement parameters, enhancing hybrid network coordination.

4 Result and discussion

The proposed Q-OptBERT achieved an accuracy and F1-score in general, and a faster convergence, in the Python-based trials with the IMDB dataset than existing quantum-enhanced baselines Q-DistilBERT-TSA, Q-DistilBERT-GWO, Q-DistilBERT-PSO, and Q-DistilBERT-GA. Due to optimized quantum parameter tuning, combined

with adaptive feature selection, it preserved the relevant semantic dependence while limiting the computational penalty of quantum parallelism. Compared to other quantum models using metaheuristic principles, it was demonstrated to be more robust to variations in text polarity and contextual nuances, illustrating generalization and efficiency for sentiment analysis problems.

4.1 Dataset collection

A popular benchmark dataset [26] for sentiment analysis and natural language processing (NLP) applications is the IMDB Dataset of 50 K Movie Reviews. Each of the 50,000 movie reviews is labelled as either positive or negative sentiment and is split equally between 25,000 for training and 25,000 for testing. Binary sentiment categorization is made possible by the dataset, which supports both conventional machine learning and deep learning techniques like CNN, LSTM, and transformer-based models. The length and tone of reviews vary, providing linguistic variation that is helpful for assessing text representation models. It is freely accessible on Kaggle and comes from the Stanford AI Lab.

4.2 Performance evaluation

The suggested Q-OptBERT model is clearly superior to the current quantum-optimized BERT variations, as shown by the performance comparison in Table 1. With the greatest accuracy (0.9824) and F1-score (0.9822), Q-OptBERT shown its strong capacity to detect subtleties in emotion. Its recall (0.9835) and precision (0.9810) ratings show a good ratio of false negatives to false positives. Superior classification reliability is confirmed by the AUC (0.972). Furthermore, Q-OptBERT demonstrated computational efficiency by requiring the least amount of training time (68 s). The effectiveness of Q-OptBERT's improved quantum–classical integration and hyperparameter tweaking is demonstrated by the fact that models such as Q-DistilBERT-GA and Q-DistilBERT-PSO trailed behind in both accuracy and speed.

Figure 2 displays the distribution of the review length, indicating that most text samples fall within a moderate range. The distribution of review length is balanced and does not result in any model biases towards very short or very long reviews, enabling the DistilBERT embeddings to attain equal contextual semantics across different text inputs. Figure 3 shows the distribution of sentiment across classes - namely, positive, negative, and neutral reviews - demonstrating that the reviews across classes are relatively well-balanced in distribution, which enables the model to learn uniformly from all categories, minimizes any skewness, and aids homogeneous

Table 1 Overall performance analysis of existing and proposed model

Model	Accuracy	Precision	Recall	F1-score	AUC	Training time (s)
Q-OptBERT (Proposed)	0.9824	0.9810	0.9835	0.9822	0.972	68
Q-DistilBERT-TSA	0.9647	0.9631	0.9640	0.9635	0.968	81
Q-DistilBERT-GWO	0.9485	0.9470	0.9478	0.9473	0.955	89
Q-DistilBERT-PSO	0.9336	0.9310	0.9325	0.9317	0.944	95
Q-DistilBERT-GA	0.9152	0.9120	0.9130	0.9125	0.931	104

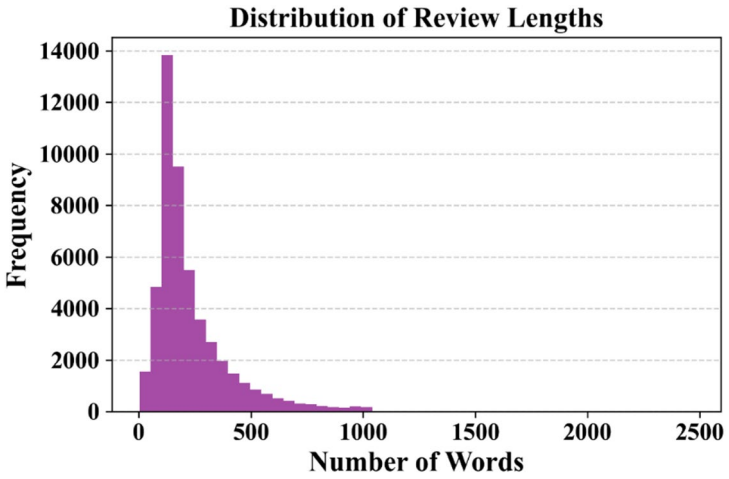
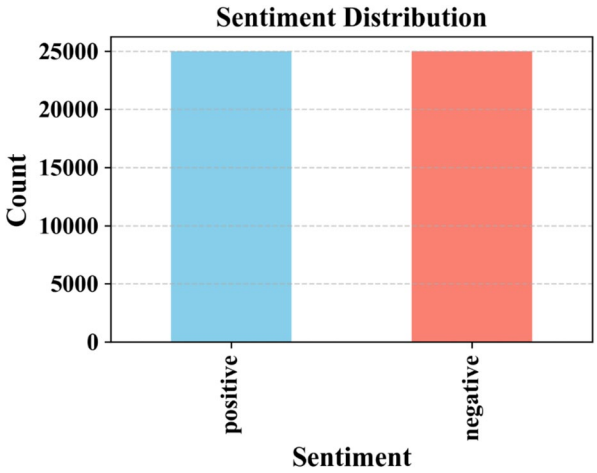


Fig. 2 Distribution of review lengths

Fig. 3 Sentiment distribution



sentiment prediction. Figure 4 indicates the training and validation accuracy and loss curves, indicating smooth convergence over epochs. The accuracy improves consistently while the loss decreases and the training and validation curves are closely aligned, indicating generalization has taken place successfully through adaptive quantum feature reduction and hybrid parameters tuned optimally. The stability of the curves confirm that Q-OptBERT alleviated overfitting and learned deeper sentiment representations. Figure 5 shows the confusion matrix and elucidates Q-OptBERT's classification precision.

The ROC curve is in Fig. 6. It depicts the true positive rate versus the false positive rate as thresholds vary. The AUC of the proposed model was 0.972 confirming discrimination between sentiment classes. The steep nature of the ROC curve near the upper left graphity line indicated high sensitivity and low false positives rates will be found. The trade-off between precision and recall in the variables assessed is

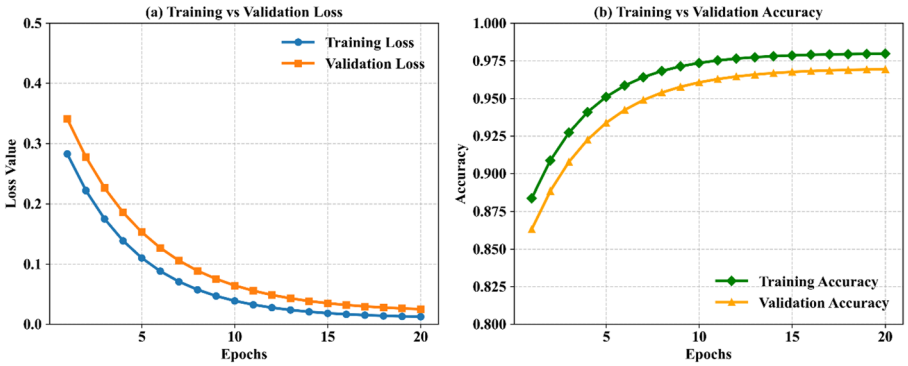


Fig. 4 Training and validation loss and accuracy

Fig. 5 Confusion matrix

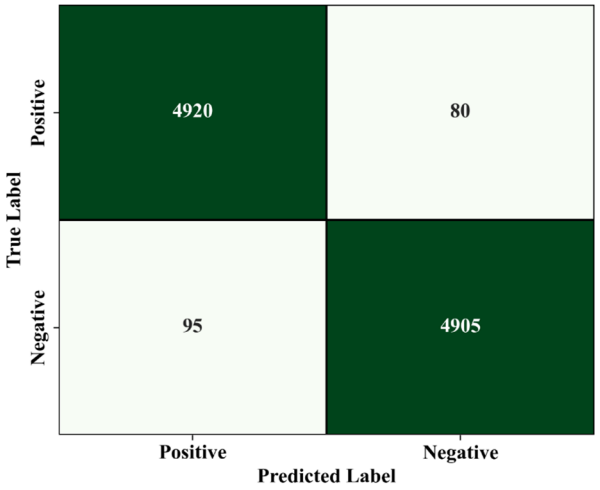
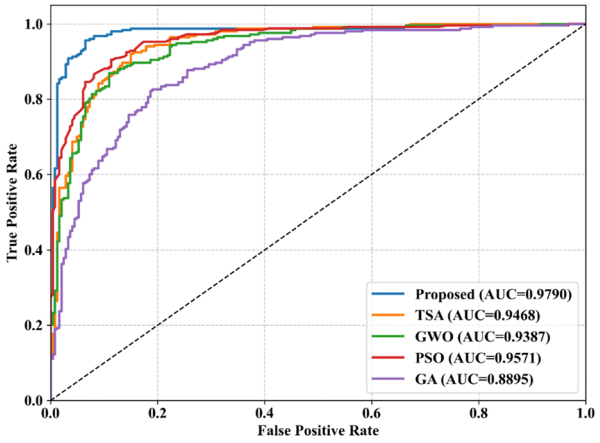


Fig. 6 ROC curve



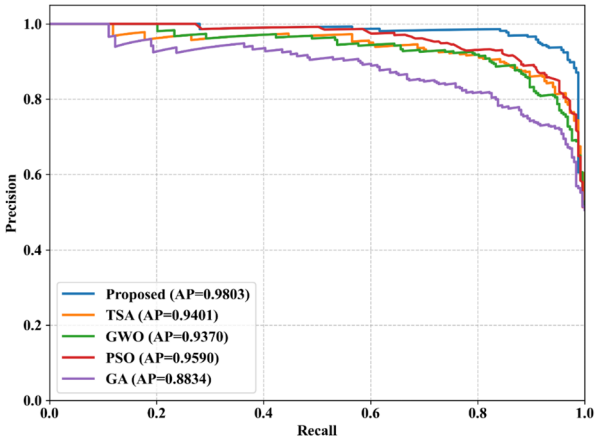


Fig. 7 PR curve

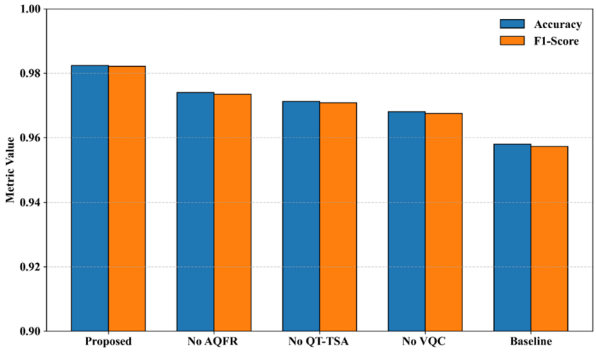


Fig. 8 Ablation study

shown by the Precision–Recall (PR) curve in Fig. 7. The Q-OptBERT PR curve can reliably retain precision while also identifying unusual or confusing attitudes since it stays high and stable over reduced recall values. This suggests that Q-OptBERT is more effective than baseline models at addressing class imbalance.

Figure 8 demonstrates the ablation study and assessment of each module associated with the quantum optimization methodology. Each configuration with satellite or TSA-based optimization removed had detrimental effects in performance, which indicates to us that all modules are capable of contributing to the final accuracy of our classifiers, including the importance of DistilBERT embeddings and adaptive quantum optimization. Figure 9 shows the attention map highlighting the sentiment-bearing words or phrases produced from higher attention weights. This interpretability layer is also an effective way of revealing that the Q-OptBERT can attend to tokens that could be considered contextually important in order to facilitate transparency of opinion as well

Figure 10 displays the class-wise evaluation metrics of the positive, negative and neutral sentiment classes with metrics encompassing accuracy, precision, recall, and F1 score. Q-OptBERT is seen to generate a high consistent metric across all classes, with very little variation between sentiment-class metrics. This would suggest that the model is robust in identifying contextual polarity and does not demonstrate bias

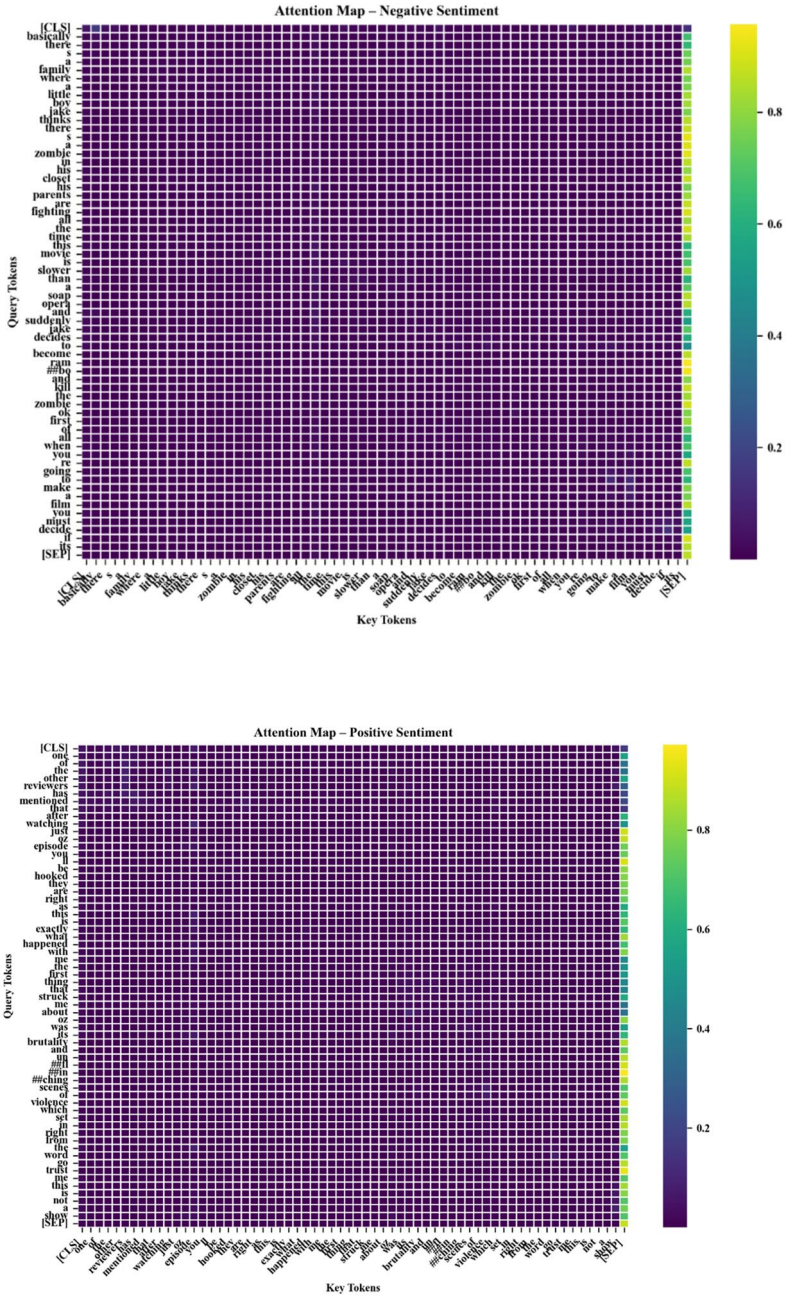


Fig. 9 Attention map

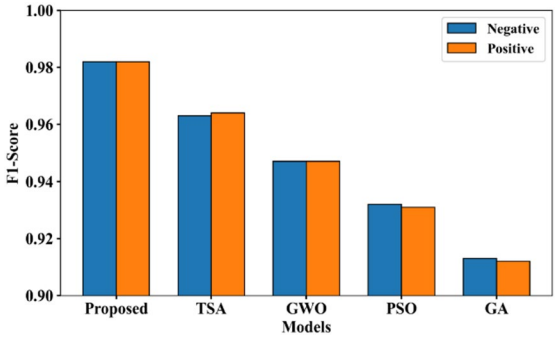


Fig. 10 Class-wise metrics

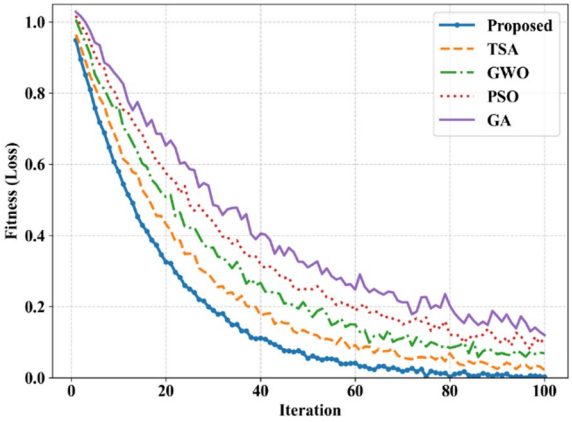


Fig. 11 Convergence analysis

toward a sentiment-type. The subtle difference in performance between positive and negative class suggests strong behavior to identify emotionally-typical expressions, while the fine-tuned identification of neutral reviews would strengthen robustness to semantics.

Figure 11 presents an analysis of convergence, which demonstrates the transience of loss and accuracy over epochs during training. Q-OptBERT exhibits a high degree of rapid convergence with considerably fewer iterations of training than that of the other models, demonstrating stability and efficiency of the hybrid quantum-classical learning mechanism. The QT-TSA (Quantum-Tuned Tunicate Swarm Algorithm) introduced in this study is a hybrid approach that both enables increased optimization efficiency and prevents overfitting during the training phase, as evidenced through smooth and stable training curves. In comparison to traditional optimization-based techniques, Q-OptBERT was able to achieve a lower final test loss with higher steady-state accuracy, establishing that the tuning differential of the Q-OptBERT model architecture is of added benefit to performance.

5 Conclusion and future scope

Q-OptBERT is a new hybrid quantum and deep learning framework designed to improve sentiment analysis with Quantum Optimization. It has the added capabilities of DistilBERT for contextual embeddings, the Variational Quantum Circuit (VQC) for representation of nonlinear relationships among linguistic variables, the Adaptive Quantum Feature Reduction (AQFR) method for dimensionality reduction, which strikes the ideal balance between semantic richness and computational efficiency. A new algorithm called the Quantum Tuned Tunicate Swarm Algorithm (QT-TSA) improves hyperparameter optimization speed, diversity and robustness of the overall model. Though this study makes contributions to the field of sentiment analysis through promising results, several limitations were identified: the use of quantum simulators instead of quantum hardware, the limitations of the quantum encoding process based on the number of qubits required for embedding dimensionality means scalability is restricted; and while there are computational gains through the hybrid training process, those gains are dependent on classical acceleration and true quantum advantage is limited by existing hardware capabilities. The limitations of QT-TSA present several possible avenues for future research. The first is to deploy Q-OptBERT on real noisy intermediate-scale quantum (NISQ) computers to more accurately evaluate its robustness, scaling, and tolerance to error in a more practical setting. The second is to extend the framework to include more complicated NLP tasks (for example, aspect-based sentiment analysis, emotion recognition, stance detection, and sarcasm detection), which will provide further evidence of the effectiveness of quantum entanglement for accurately capturing the subtleties of linguistic dependencies. Third, to make large-scale text processing more feasible, future work may explore methods for improving the efficiency of quantum encoding (reducing the number of qubit and the depth of the circuit). Fourth, by integrating QT-TSA with approaches such as reinforcement learning or evolutionary algorithms, researchers may enhance adaptive hyperparameter tuning capabilities. Finally, the creation of quantum interpretability mechanisms will tremendously improve the transparency and assurance in quantum-enhanced NLP systems, allowing for the interpretation of how quantum information influences the way in which a quantum state represents language.

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Data availability Authors are working on implementing the same using real world data with appropriate permissions.

Declarations

Conflict of interest The authors declare that we have no competing interest.

Ethical approval and consent to participate Not Applicable.

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